Assignment → 5

Q.1 Define a tree and explain its basic terminology with a suitable example. Also explain the properties of a binary tree.

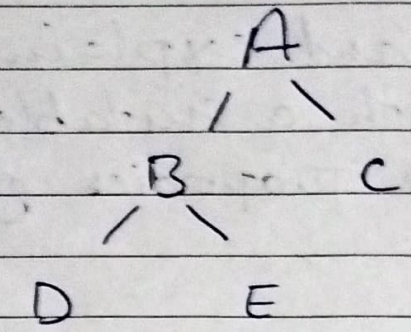
Ans Tree:
A non-linear data structure consisting of nodes connected by edges.

Terminology:-
- Root: Topmost Node.
- Parent: Node having children.
- Child: Node below a Parent.
- Leaf: Node having no children.
- Sibling: Nodes with the same Parent.
- Height: Longest path from root to leaf.

Binary Tree Properties:-
1. Each node has at most two children.
2. Maximum nodes at level $L = 2^L$.
3. Maximum nodes at height $H = 2^{H+1} - 1$
4. A binary tree has no cycles.

Q.2 Explain binary tree traversal techniques: Preorder, Inorder, and Postorder. Construct a binary tree and demonstrate all three travels.

Ans Traversal means visiting each and every node present in a tree.



→ Preorder (~~Left~~ Root, Left, Right): A, B, D, E, C.

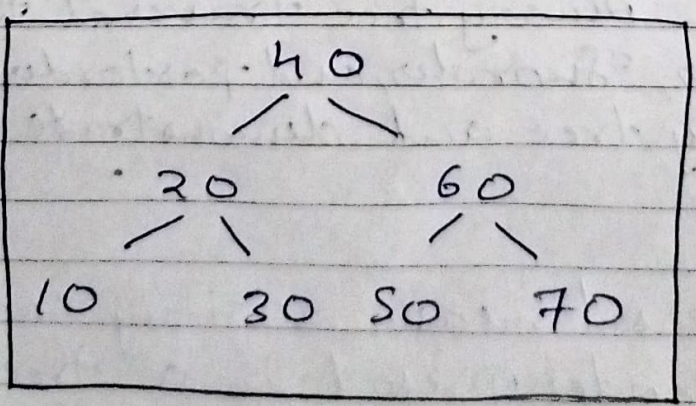
→ Inorder (Left, Root, Right): D, B, E, A, C

→ Postorder (Left, Right, Root): D, E, B, C, A

Q.3 What is Binary Search Tree (BST)? Construct a BST for the following values and show the tree after each insertion: 40, 20, 60, 10, 30, 50, 70.

Ans BST: A Binary Search Tree where left child values are smaller and right child values are greater that the parent.

Final BST:-



Q.4 Explain AVL trees and the need for balancing. Insert the following values into an AVL tree & show the rotations required : 30, 20, 10, 25, 28.

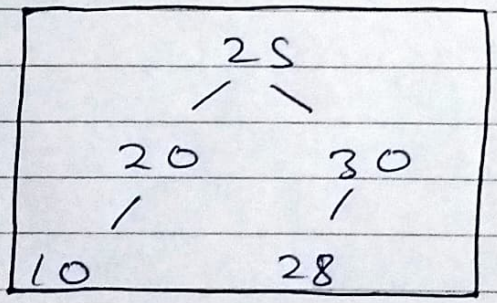
Ans AVL Tree : A self-balancing BST where the balance factor is $-1, 0$ or $+1$.

- Need : Maintains balanced height and provide $O(\log n)$ operations.

→ Given : 30, 20, 10, 25, 28.

- Insert : 30, 20, 10 → LL Rotation (Right Rotate)
- Insert : 25 → No Rotation
- Insert : 28 → RL Rotation

Final Tree :



Q.5 Explain the representation of graphs using an adjacency matrix and adjacency list. Demostrate Breadth-First Search (BFS) and Depth-First Search (DFS) on a suitable graph.

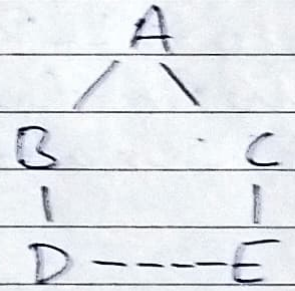
Ans Graph: A graph is a collection of vertices and edges.

Representation:

1. Adjacency Matrix: Uses a 2D matrix to represent edges
2. Adjacency List: Stores adjacent vertices in a list.

Example:-

→ Adjacency Matrix



→ Adjacency List:-

A → B, C
B → A, D
C → A, E
D → B, E
E → C, D

- BFS: Visit nodes level-wise using a queue
  → Output: A, B, C, D, E
- DFS → Visit one path deeply before backtracking
  → Output: A, B, D, E, C

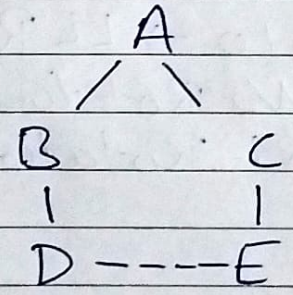
Ans Graph: A graph is a collection of vertices and edges.

Representation:

1. Adjacency Metric: Uses a 2D matrix to represent edges.
2. Adjacency List: Stores adjacent vertices in a list.

Example:-

→ Adjacency Metrix



→ Adjacency List:-

A → B, C
B → A, D
C → A, E
D → B, E
E → C, D

- BFS: Visit nodes level-wise using a queue
  → Output: A, B, C, D, E
- DFS → Visit one path deeply before backtracking
  → Output: A, B, D, E, C.

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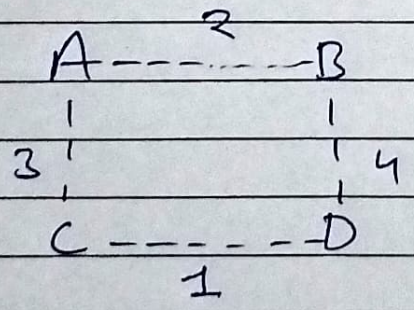
Q.6 What is a Minimum Spanning Tree (MST)? Explain Prim's algorithm with a suitable weighted graph. Also describe one real-life application of MST.

Ans MST: A spanning tree connecting all vertices with minimum total edge weight.

Prim's Algorithm:-

1. Start from any vertex.
2. Select the minimum-weight connecting edge.
3. Add the new vertex.
4. Repeat until all vertices are connected.

Example:-



Mst Edges: A - B = 2, A - C = 3, C - D = 1
Total cost ⇒ 6.

Application: Used in designing electrical networks and communication cables at minimum cost.